

Fig. 5: Example of one-time pad encryption using modulo 26

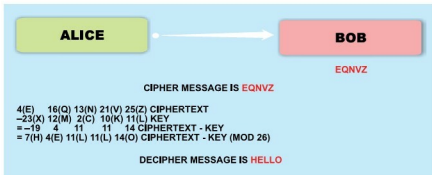


Fig. 6: Example of one-time pad decryption using modulo 26

if you try to encrypt the message 'INDIA' with pad 'XMCKL', you will get the encrypted text as GZFTL. Here the strength of one-time pads is also in providing more than one meaningful text which can then put a sniffer into confusion regarding the key used.

For example, considering the string 'HELLO' being encrypted using the pad 'XMCKL', the output obtained is 'EQNVZ'. A sniffer who is assumed to have infinite computing power, and finds that the key 'XMCKL' produces the plaintext 'HELLO', will also find that the key 'TQURI' produces the plaintext 'LATER'—an equally plausible message—for the encrypted output of 'EQNVZ'.

The following actions are required to organise the pad list:

1. Import: Save other users' pads into your folder
 2. Export: Save your own pads on a carrier to deliver it to another user
 3. Delete: Remove a pad from your list
 4. Move: Remove a pad from your list and move it to another location
 5. Change Exp: Change properties of your key to 'expired' or back to 'valid'
- If the key has used over 90 per

cent of its bytes, the program should ask to mark the key as 'expired.' Otherwise, one can use the key until no more bytes are available. If the key is set to 'expired' before encryption, this is noted in the segment info of the encrypted message. In this case, if the receiver of the message decrypts the message, the key that was imported in his list will be changed automatically to 'expired.' Always inform the person that received your exported key when it is no longer in use and can be deleted or set to 'expired.' Expired keys cannot be used any longer for encryption, but are still available for decrypting.

In some cases, if encrypted information is stored for a longer time, it can be useful to change the key to 'expired' instead of deleting it. However, if it's no longer used, it is advised to delete the key. One can change the status of the key with the 'Change Exp' button. This is only an indication. If there are no more free bytes, one can't encrypt with the key, even if its status is changed to 'valid.'

In the one-time-pad list or the selection list, one can check the properties of the selected key. These are the possibilities:

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